

Theory Assignment 1

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Q1.E-learning Course Page

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>E-Learning Course Page</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <!-- Header section containing the course title and navigation -->
  <header>
    <h1>Web Basics Bootcamp</h1>
    <p>Learn HTML, CSS, and JavaScript step by step.</p>

    <!-- Semantic navigation bar with module links -->
    <nav aria-label="Main navigation">
      <ul class="nav-list">
        <li><a href="#home">Home</a></li>
        <li><a href="#modules">Modules</a></li>
        <li><a href="#videos">Videos</a></li>
        <li><a href="#resources">Resources</a></li>
        <li><a href="#feedback">Feedback</a></li>
      </ul>
    </nav>

    <!-- Main page content -->
    <main>
```

```
<!-- Section 1: Course overview and module list -->
```

```
<section id="home" class="card">
```

```
  <h2>Course Overview</h2>
```

```
  <article>
```

```
    <h3>What You Will Learn</h3>
```

```
    <p>
```

This beginner course helps you build strong web development fundamentals.

You will practice structure, styling, and basic interactivity.

```
    </p>
```

```
    <ul>
```

```
      <li>Write semantic HTML pages</li>
```

```
      <li>Style pages with CSS and Flexbox</li>
```

```
      <li>Create simple dynamic behavior using JavaScript</li>
```

```
    </ul>
```

```
  </article>
```

```
</section>
```

```
<!-- Section 2: Modules in ordered list format -->
```

```
<section id="modules" class="card">
```

```
  <h2>Course Modules</h2>
```

```
  <article>
```

```
    <h3>Learning Path</h3>
```

```
    <ol>
```

```
      <li>Introduction to HTML</li>
```

```
      <li>CSS Layout and Styling</li>
```

```
      <li>Responsive Web Design Basics</li>
```

```
      <li>JavaScript Fundamentals</li>
```

```
    </ol>
```

```
  </article>
```

```
</section>
```

```
<!-- Section 3: Embedded video lesson -->
```

```
<section id="videos" class="card">
  <h2>Featured Video Lesson</h2>
  <article>
    <h3>HTML Crash Course</h3>
    <div class="video-wrapper">
      <iframe
        src="https://www.youtube.com/embed/qz0aGYrrlhU"
        title="YouTube video player"
        allow="accelerometer; autoplay; clipboard-write;
encrypted-media; gyroscope; picture-in-picture"
        allowfullscreen>
      </iframe>
    </div>
  </article>
</section>
```

<!-- Section 4: Downloadable resource links -->

```
<section id="resources" class="card">
  <h2>Downloadable Resources</h2>
  <article>
    <h3>Study Material</h3>
    <ul>
      <li><a href="#">HTML Quick Guide (PDF)</a></li>
      <li><a href="#">CSS Practice Notes</a></li>
      <li><a href="#">JavaScript Basics Worksheet</a></li>
    </ul>
  </article>
</section>
```

<!-- Section 5: Basic feedback form (no JavaScript) -->

```
<section id="feedback" class="card">
  <h2>Course Feedback</h2>
  <form action="#" method="post">
    <label for="name">Name</label>
```

```
        <input type="text" id="name" name="name" placeholder="Enter your
name" required>
```

```
        <label for="email">Email</label>
```

```
        <input type="email" id="email" name="email" placeholder="Enter
your email" required>
```

```
        <label for="message">Message</label>
```

```
        <textarea id="message" name="message" rows="5" placeholder="Share
your feedback..." required></textarea>
```

```
        <button type="submit">Submit Feedback</button>
```

```
    </form>
```

```
</section>
```

```
</main>
```

```
<!-- Footer information -->
```

```
<footer>
```

```
    <p>&copy; 2026 Web Basics Bootcamp. All rights reserved.</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

style.css

/* Basic reset for consistent spacing across browsers */

```
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}
```

/* Page-level styling for readable typography and soft background */

```
body {  
  font-family: Arial, sans-serif;  
  line-height: 1.6;  
  background-color: #f4f7fb;  
  color: #1f2937;  
}
```

/* Header styling to create clear visual separation */

```
header {  
  background-color: #1d4ed8;  
  color: #ffffff;  
  padding: 1.5rem 1rem;  
  text-align: center;  
}
```

```
header p {  
  margin-top: 0.4rem;  
}
```

/* Flexbox navbar for horizontal links on wider screens */

```
nav {  
  margin-top: 1rem;  
}
```

```
.nav-list {  
    list-style: none;  
    display: flex;  
    justify-content: center;  
    gap: 1rem;  
    flex-wrap: wrap;  
}
```

```
.nav-list a {  
    color: #ffffff;  
    text-decoration: none;  
    padding: 0.5rem 0.8rem;  
    border-radius: 6px;  
    transition: background-color 0.2s ease;  
}
```

```
.nav-list a:hover {  
    background-color: rgba(255, 255, 255, 0.2);  
}
```

```
/* Main content container */
```

```
main {  
    max-width: 900px;  
    margin: 1.5rem auto;  
    padding: 0 1rem;  
}
```

```
/* Card style used by each section to keep layout clean */
```

```
.card {  
    background-color: #ffffff;  
    padding: 1.2rem;  
    border-radius: 10px;  
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);  
}
```

```
        margin-bottom: 1.2rem;
    }

h2 {
    margin-bottom: 0.8rem;
    color: #1e3a8a;
}

h3 {
    margin-bottom: 0.6rem;
}

ul,
ol {
    margin-left: 1.2rem;
    margin-top: 0.5rem;
}

/* Responsive YouTube embed container */
.video-wrapper {
    position: relative;
    width: 100%;
    padding-top: 56.25%;
    border-radius: 8px;
    overflow: hidden;
    margin-top: 0.8rem;
}

.video-wrapper iframe {
    position: absolute;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
}
```

```
border: 0;
}

/* Basic form styling */
form {
  display: flex;
  flex-direction: column;
  gap: 0.6rem;
}

input,
textarea {
  padding: 0.6rem;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  font-size: 1rem;
}

button {
  width: fit-content;
  padding: 0.7rem 1rem;
  border: none;
  border-radius: 6px;
  background-color: #2563eb;
  color: #ffffff;
  cursor: pointer;
}

button:hover {
  background-color: #1e40af;
}

footer {
  text-align: center;
```



```
padding: 1rem;
background-color: #111827;
color: #ffffff;
margin-top: 1rem;
}

/* On small screens, stack nav links vertically for readability */
@media (max-width: 600px) {
    .nav-list {
        flex-direction: column;
        align-items: center;
    }

    .card {
        padding: 1rem;
    }
}
```

Web Basics Bootcamp

Learn HTML, CSS, and JavaScript step by step.

[Home](#) [Modules](#) [Videos](#) [Resources](#) [Feedback](#)

Course Overview

What You Will Learn

This beginner course helps you build strong web development fundamentals. You will practice structure, styling, and basic interactivity.

- Write semantic HTML pages
- Style pages with CSS and Flexbox
- Create simple dynamic behavior using JavaScript

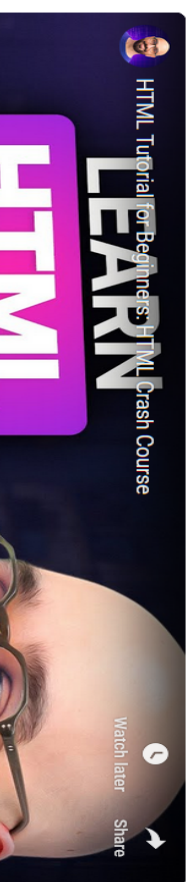
Course Modules

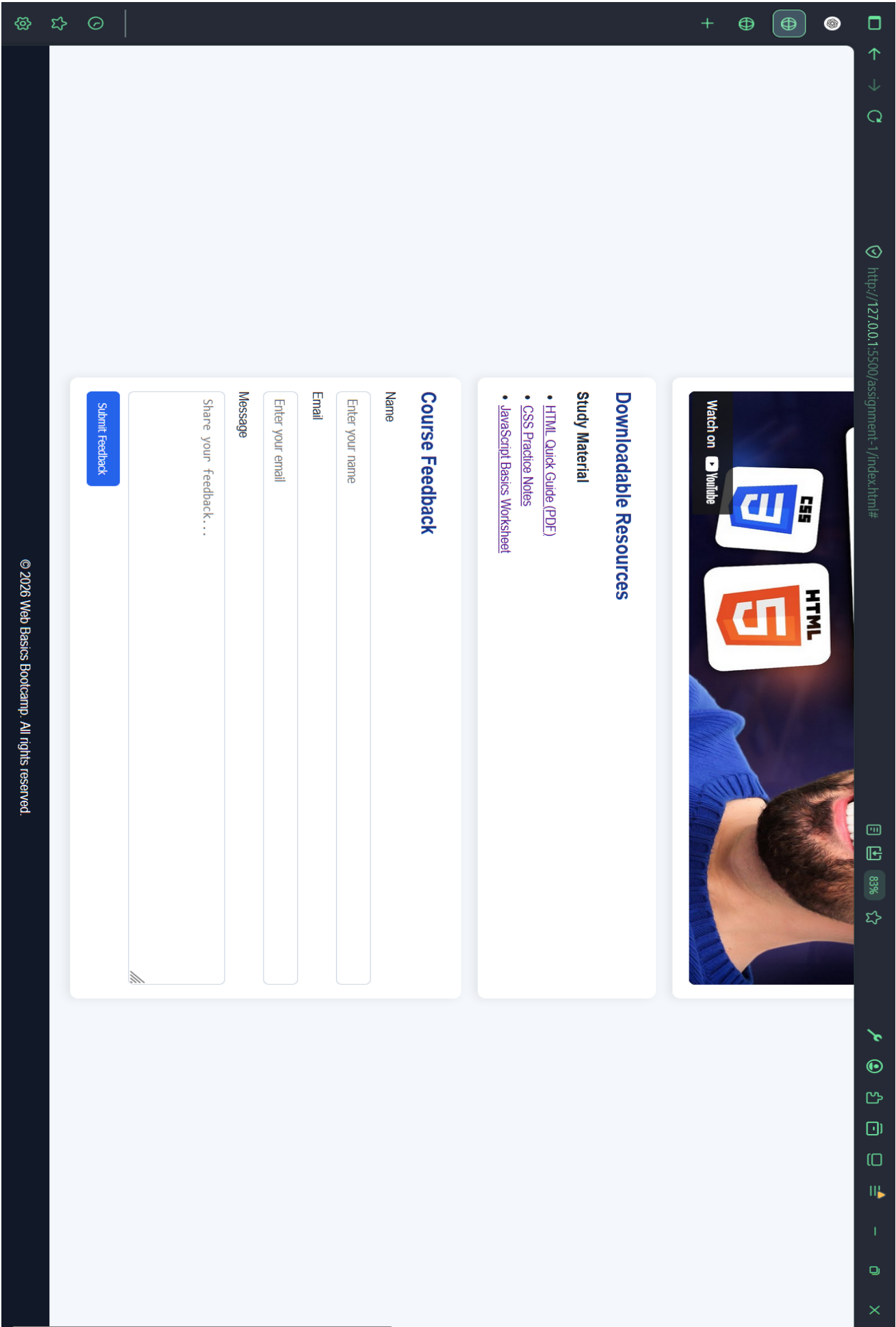
Learning Path

1. Introduction to HTML
2. CSS Layout and Styling
3. Responsive Web Design Basics
4. JavaScript Fundamentals

Featured Video Lesson

HTML Crash Course





Q2. Dynamic Feedback Form

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Feedback Form</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="form-page">
    <section class="form-card">
      <h1>Course Feedback Form</h1>
      <p>Please fill out all fields before submitting your feedback.</p>

      <!-- Form uses novalidate so custom JavaScript validation can handle
messages -->
      <form id="feedbackForm" novalidate>
        <div class="field-group">
          <label for="name">Name</label>
          <input type="text" id="name" name="name" placeholder="Enter
your name">
          <small class="error" id="nameError"></small>
        </div>

        <div class="field-group">
          <label for="email">Email</label>
          <input type="email" id="email" name="email"
placeholder="Enter your email">
          <small class="error" id="emailError"></small>
        </div>

        <div class="field-group">
```

```
<label for="rating">Rating (1 to 5)</label>
<select id="rating" name="rating">
    <option value="">Select rating</option>
    <option value="1">1</option>
    <option value="2">2</option>
    <option value="3">3</option>
    <option value="4">4</option>
    <option value="5">5</option>
</select>
<small class="error" id="ratingError"></small>
</div>

<div class="field-group">
    <label for="comments">Comments</label>
    <textarea id="comments" name="comments" rows="4"
placeholder="Write your comments..."></textarea>
    <small class="error" id="commentsError"></small>
</div>

<button type="submit">Submit Feedback</button>
<p id="successMessage" class="success"></p>
</form>
</section>
</main>

<script src="script.js"></script>
</body>
</html>
```

style.css

```
/* Center the form card on the page */
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background-color: #f0f4f8;
    color: #1f2937;
}

.form-page {
    min-height: 100vh;
    display: flex;
    align-items: center;
    justify-content: center;
    padding: 1rem;
}

/* Simple card layout for beginner-friendly styling */
.form-card {
    width: 100%;
    max-width: 450px;
    background-color: #ffffff;
    padding: 1.5rem;
    border-radius: 10px;
    box-shadow: 0 4px 14px rgba(0, 0, 0, 0.1);
}

h1 {
    margin-top: 0;
    margin-bottom: 0.5rem;
    color: #1e3a8a;
}
```

```
p {  
    margin-top: 0;  
    margin-bottom: 1rem;  
}
```

```
.field-group {  
    margin-bottom: 1rem;  
}
```

```
label {  
    display: block;  
    margin-bottom: 0.4rem;  
    font-weight: bold;  
}
```

```
input,  
select,  
textarea {  
    width: 100%;  
    padding: 0.6rem;  
    border: 1px solid #cbd5e1;  
    border-radius: 6px;  
    font-size: 1rem;  
}
```

```
button {  
    width: 100%;  
    padding: 0.7rem;  
    border: none;  
    border-radius: 6px;  
    background-color: #2563eb;  
    color: #ffffff;  
    font-size: 1rem;  
    cursor: pointer;
```

```
}
```

```
button:hover {  
    background-color: #1e40af;  
}
```

```
/* Error text shown under each input field */
```

```
.error {  
    display: block;  
    margin-top: 0.3rem;  
    color: #dc2626;  
    min-height: 1rem;  
}
```

```
/* Success message shown after valid submission */
```

```
.success {  
    margin-top: 1rem;  
    color: #15803d;  
    font-weight: bold;  
    min-height: 1.2rem;  
}
```


script.js

```
// Get form and field references once so we can reuse them in functions.
```

```
const form = document.getElementById("feedbackForm");
```

```
const nameInput = document.getElementById("name");
```

```
const emailInput = document.getElementById("email");
```

```
const ratingInput = document.getElementById("rating");
```

```
const commentsInput = document.getElementById("comments");
```

```
const nameError = document.getElementById("nameError");
```

```
const emailError = document.getElementById("emailError");
```

```
const ratingError = document.getElementById("ratingError");
```

```
const commentsError = document.getElementById("commentsError");
```

```
const successMessage = document.getElementById("successMessage");
```

```
// Validation function for name: must not be empty.
```

```
function validateName() {
```

```
    const nameValue = nameInput.value.trim();
```

```
    if (nameValue === "") {
```

```
        nameError.textContent = "Name is required.";
```

```
        return false;
```

```
    }
```

```
    nameError.textContent = "";
```

```
    return true;
```

```
}
```

```
// Validation function for email using a simple regex format check.
```

```
function validateEmail() {
```

```
    const emailValue = emailInput.value.trim();
```

```
    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
    if (emailValue === "") {
```

```
        emailError.textContent = "Email is required.";
        return false;
    }

    if (!emailPattern.test(emailValue)) {
        emailError.textContent = "Please enter a valid email address.";
        return false;
    }

    emailError.textContent = "";
    return true;
}

// Validation function for rating: must be selected and between 1 and 5.
function validateRating() {
    const ratingValue = Number(ratingInput.value);

    if (ratingInput.value === "") {
        ratingError.textContent = "Rating is required.";
        return false;
    }

    if (ratingValue < 1 || ratingValue > 5) {
        ratingError.textContent = "Rating must be between 1 and 5.";
        return false;
    }

    ratingError.textContent = "";
    return true;
}

// Validation function for comments: must not be empty.
function validateComments() {
    const commentsValue = commentsInput.value.trim();
```

```

    if (commentsValue === "") {
        commentsError.textContent = "Comments are required.";
        return false;
    }

    commentsError.textContent = "";
    return true;
}

// Run all validation functions and return true only when all are valid.
function validateForm() {
    const isNameValid = validateName();
    const isEmailValid = validateEmail();
    const isRatingValid = validateRating();
    const isCommentsValid = validateComments();

    return isNameValid && isEmailValid && isRatingValid && isCommentsValid;
}

// Handle submit event: prevent default submit, show errors or success message.
form.addEventListener("submit", function (event) {
    event.preventDefault();
    successMessage.textContent = "";

    if (validateForm()) {
        successMessage.textContent = "Feedback submitted successfully!";
        form.reset();
    }
});

// Optional live validation: checks each field after the user leaves the input.
nameInput.addEventListener("blur", validateName);
emailInput.addEventListener("blur", validateEmail);

```

```
ratingInput.addEventListener("blur", validateRating);
commentsInput.addEventListener("blur", validateComments);
```

Course Feedback Form

Please fill out all fields before submitting your feedback.

Name

Email

Rating (1 to 5)

Select rating

Comments

Write your comments...

Submit Feedback

Course Feedback Form

Please fill out all fields before submitting your feedback.

Name

Email

Rating (1 to 5)

4

Select rating

1

2

3

4

5

Submit Feedback

Course Feedback Form

Please fill out all fields before submitting your feedback.

Name

Email

Rating (1 to 5)

4

Comments

great learning experience

Submit Feedback

Course Feedback Form

Please fill out all fields before submitting your feedback.

Name

Name is required.

Email

Email is required.

Rating (1 to 5)

Select rating

Rating is required.

Comments

Write your comments...

Comments are required.

Submit Feedback